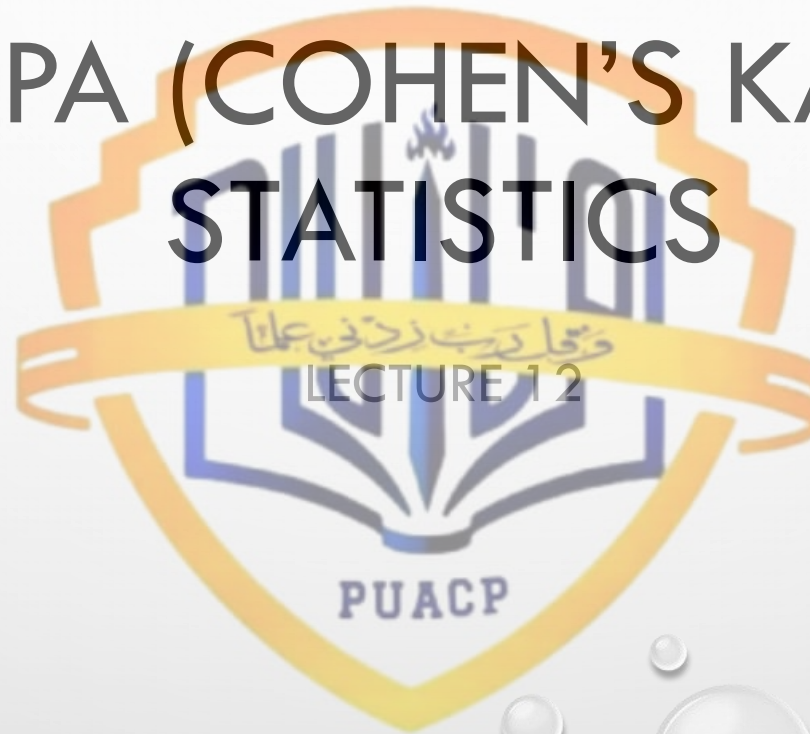




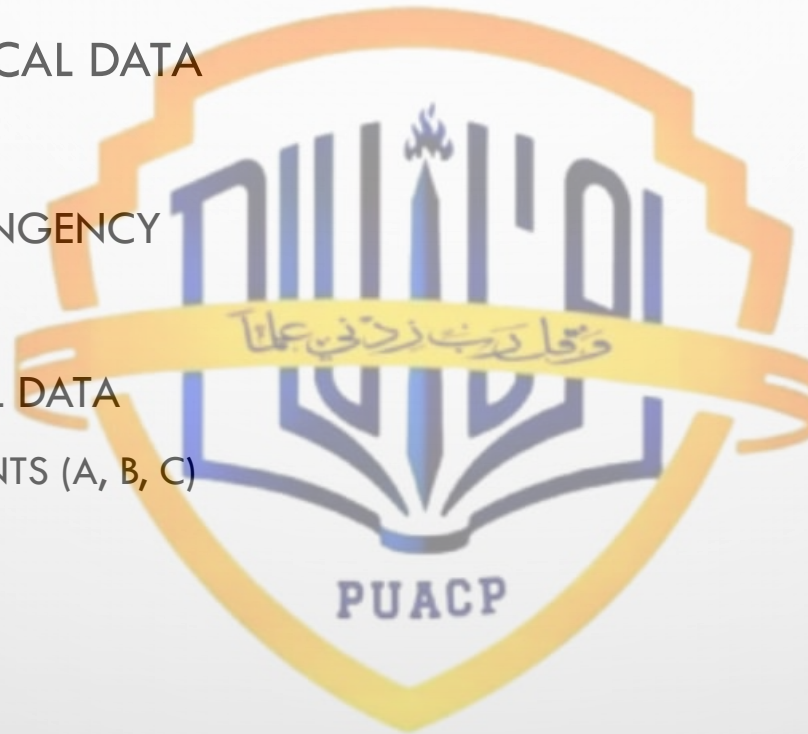
KAPPA (COHEN'S KAPPA) STATISTICS



LECTURE 12

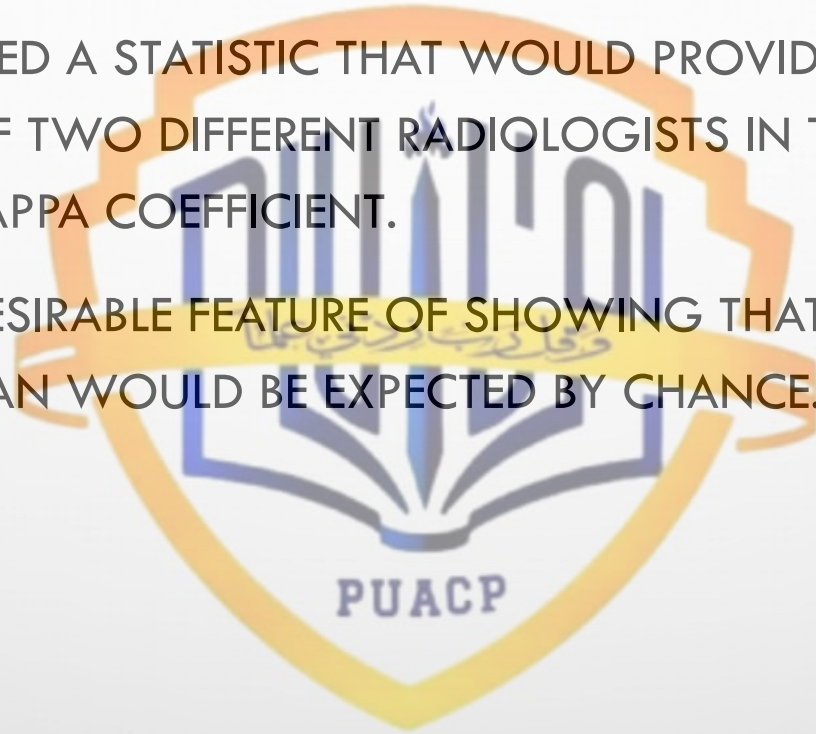
MEASURE OF ASSOCIATIONS FOR CATEGORICAL DATA

- FOR NOMINAL CATEGORICAL DATA
 - PHI- COEFFICIENT
 - COEFFICIENT OF CONTINGENCY
 - CRAMER'S V
- FOR ORDINAL CATEGORICAL DATA
 - KENDALL'S TAU COEFFICIENTS (A, B, C)
 - SOMERS' D



KAPPA STATISTIC

- IN 1960, COHEN PROPOSED A STATISTIC THAT WOULD PROVIDE A MEASURE OF AGREEMENT BETWEEN THE RATINGS OF TWO DIFFERENT RADIOLOGISTS IN THE INTERPRETATION OF X-RAYS. HE CALLED IT THE KAPPA COEFFICIENT.
- THIS MEASURE HAS THE DESIRABLE FEATURE OF SHOWING THAT HOW MUCH MORE AGREEMENT IS THERE , THAN WOULD BE EXPECTED BY CHANCE.



- IN CLINICAL TRIAL, MEDICAL RESEARCH AND MANY OTHER SITUATIONS, WE OFTEN HAVE A SITUATION WHERE TWO DIFFERENT MEASURES/ASSESSMENTS ARE PERFORMED ON THE SAME SAMPLE, SAME PATIENT, SAME IMAGE,... THE AGREEMENT NEEDS TO BE CALCULATED AS A SUMMARY STATISTICS.

EXAMPLE

- IN A CLINICAL TRIAL WITH IMAGING ASSESSMENT, THE SAME IMAGE (FOR EXAMPLE, CT SCAN,...) CAN BE READ BY DIFFERENT READERS
- FOR DISEASE DIAGNOSIS, A NEW DIAGNOSTIC TOOL (WITH ADVANTAGE OF LESS INVASIVE OR EASIER TO IMPLEMENT) COULD BE COMPARED TO AN ESTABLISHED DIAGNOSTIC TOOL... TYPICALLY, THE OUTCOME MEASURE IS DICHOTOMOUS (E.G., DISEASE VS NO DISEASE, POSITIVE VS. NEGATIVE...).

- DEPENDING ON WHETHER OR NOT THE MEASUREMENT IS CONTINUOUS OR CATEGORICAL, THE AGREEMENT STATISTICS COULD BE DIFFERENT.
- SPECIFICALLY FOR CATEGORICAL ASSESSMENT, THERE ARE MANY EXAMPLES WHERE THE AGREEMENT STATISTICS IS NEEDED.
- FOR EXAMPLE, FOR COMPARING THE ASSESSMENT FROM TWO DIFFERENT READERS, WE WOULD CALCULATE KAPPA, OVERALL PERCENT AGREEMENT, POSITIVE PERCENT AGREEMENT, AND NEGATIVE PERCENT AGREEMENT.
- KAPPA STATISTIC(K) IS A MEASURE OF AGREEMENT BETWEEN TWO SOURCES, WHICH IS MEASURED ON A BINARY SCALE (I.E. CONDITION PRESENT/ABSENT).
- K STATISTIC CAN TAKE VALUES BETWEEN 0 AND 1.

KAPPA STATISTIC

$$K = \frac{P_o - P_e}{1 - P_e}$$

WHERE

P_o = OBSERVE PROPORTION OF AGREEMENT

P_e = EXPECTED PROPORTION OF AGREEMENT UNDER THE ASSUMPTION OF INDEPENDENCE

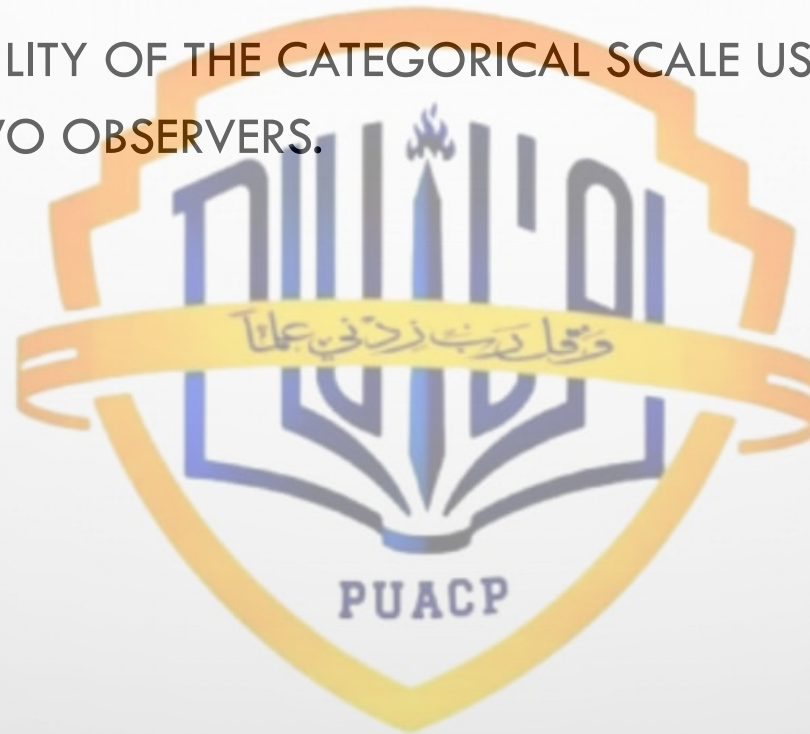
LANDIS AND KOCH (1977) PROVIDED THE GUIDELINES FOR EVALUATING THE KAPPA STATISTIC. THESE VALUES ARE USED AS BENCHMARK FOR EVALUATING THE VALUE OF KAPPA COEFFICIENT.

Value of Kappa	Level of Agreement	% of Data that are Reliable
0-.20	None	0-4%
.21-.39	Minimal	4-15%
.40-.59	Weak	15-35%
.60-.79	Moderate	35-63%
.80-.90	Strong	64-81%
Above .90	Almost Perfect	82-100%



IMPORTANCE AND SHORTCOMING

- **IMPORTANCE:** THE RELIABILITY OF THE CATEGORICAL SCALE USUALLY BE EVALUATING BY AGREEMENT BETWEEN TWO OBSERVERS.
- **SHORTCOMING:**



EXAMPLE

- A DIET QUESTIONNAIRE WAS FILLED BY 537 FEMALES ON TWO DIFFERENT OCCASIONS, SEVERAL MONTHS APART REGARDING BEEF CONSUMPTION.

Survey1\ Survey 2	=1 serving\ week	>1 serving\ week	Total
=1 serving\ week	136	92	228
>1 serving\ week	69	240	309
Total	205	332	537

- CALCULATE THE KAPPA STATISTIC TO CHECK THE AGREEMENT BETWEEN TWO SURVEYS.

EXAMPLE

- AS FROM THE FORMULA WE CAN SEE THAT OBSERVED AND EXPECTED COUNT OF AGREEMENT ARE NEEDED. THE EXPECTED COUNTS WILL BE CALCULATED AS WE HAVE ALREADY DONE FOR CHI- SQUARE STATISTICS

Survey1\ Survey 2	=1 serving\ week	>1 serving\ week	Total	
=1 serving\ week	136 (87)	92 (141)	228	
>1 serving\ week	69 (118)	240 (191)	309	
Total	205	332	537	

CALCULATIONS

$$K = \frac{P_o - P_e}{1 - P_e}$$

$P_o = (137 + 240) / 537 = 0.70$ (Observed proportion of agreement)

$P_e = (87 + 191) / 537 = 0.52$ (Expected proportion of agreement)

$$K = \frac{0.70 - 0.52}{1 - 0.52} = 0.375 = 37.5\%$$

Interpretation:

Since, $k = 37.5\%$, therefore according to scale suggested by Ladina and Koch (1977) there is fair agreement between two surveys.

PRACTICE QUESTION

- THE DATA REGARDING THE AGREEMENT ABOUT THE SEVERITY OF DISEASE FOR THE FIRST AND SECOND EXAMINATION FOR 183 PATIENTS IS AS FOLLOWS,

1 st Examination	2 nd Examination		
	Normal	Grade 1	Grade 2
Normal	72	46	0
Grade 1	6	47	17
Grade 2	1	14	20

- CALCULATE THE AGREEMENT BETWEEN TWO EXAMINATIONS.

PRACTICE QUESTION 2

- THE JOINT RATING OF TWO PSYCHIATRISTS REGARDING 118 PATIENTS HAVE BEEN DISPLAYED IN THE FOLLOWING TABLE . CALCULATE MEASURE OF AGREEMENT BETWEEN TWO PSYCHIATRISTS.

2 nd Psychiatrists	1 st Psychiatrists				
	D1	D2	D3	D4	D4
D1	22	2	2	0	0
D2	5	7	14	0	0
D3	0	2	36	0	0
D4	0	1	14	7	0
D5	0	0	3	0	3

K= 49%